



Design Brief

FOB02-04A: A Bitopic Ligand and the Dopamine D3 Receptor

DECEMBER 2025

COMPLETED FOR

DR. NOWLAN | NSCI 405 ADVANCED MOLECULAR NEUROPHARMACOLOGY
UNC CHAPEL HILL

DESIGN TEAM

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Introduction & Background

The Makerspace project for NSCI 405 involves the design of a 3-D printed explanatory model of how a new drug of clinical interest works at its receptor target.

Our goal is to design a model that:

1. Demonstrates a newer, novel mechanism of action, like the kinds of allosteric interactions we will focus on in class
2. Illustrates a conformational change in the receptor
3. Can be used to explain to stakeholders how the new, novel drug works

The objective of this Design Dossier is to document all steps in the design process.



Project Description

The purpose of this project is as follows:

1. Demonstrate how the bitopic ligand FOB02-04A engages both the orthosteric pocket and an extracellular vestibular/allosteric site to stabilize the receptor's active GPCR conformation and potentially enable better therapeutic targeting in treatment of neurodegeneration.
2. Show how the D3 receptor changes conformations upon interaction with the bitopic FOB02-04A ligand and illustrate how the ligand's secondary pharmacophore targets a D3-specific allosteric "selectivity site," distinguishing D3 from other dopamine receptors and having subtype-selective activation with potentially fewer off-target effects.
3. Provide a clear communication tool to explain the function and mechanism of FOB02-04A to medicinal chemists and research scientists, and more broadly to illustrate the function of bitopic ligands at the D3 receptor in how they affect structure and signaling.

The description of the project is as follows:

1. Our group is interested in studying the D3 dopamine metabotropic receptor and its association with novel compound FOB02-04A, a bitopic ligand that binds to both

orthosteric and allosteric sites to produce a strong agonist effect. Other D3 agonists have been studied in clinical trials as treatments for the symptoms of Parkinson's Disease, a neurodegenerative disease that involves the death of dopaminergic neurons (Carnicella et al., 2014).

2. FOB02-04A has a relatively unique mechanism of action, as it is both an agonist compound and a positive allosteric modulator. This is done by connecting an agonist compound and a positive allosteric modulator with a linker region, allowing one molecule to produce strong agonist activity without requiring a separate positive allosteric modulator. More specifically, the drug's primary pharmacophore (PP) occupies the deep orthosteric pocket, while its secondary pharmacophore (SP) engages a D3-specific vestibular site; a linker connects them. The binding of the ligand at both sites shifts the receptor into a more stable active shape (Arroyo-Urea et al., 2024).
3. Producing a three dimensional model of this drug-receptor interaction would help medicinal chemists by allowing them to easily understand how FOB02-04A's mechanism of action is unique compared to orthosteric and allosteric modulators, as well as the specifics of how it physically interacts with the D3 receptor. One way our model could show the ligand-receptor interaction is by creating a model that can open up and has the binding pockets for the ligand. This model would exist in the inactive conformation without the ligand and would shift into the active conformation once the bitopic ligand is inserted. This transformation will explain how the conformations differ from one another and how the bitopic ligand puts the D3 receptor into the active conformation. Additionally, the model can show how the binding of FOB02-04A causes a conformational shift that stabilizes the ligand, illustrating that the drug drives the receptor to a stronger, and more active confirmation (Arroyo-Urea et al., 2024). The model can also serve to compare D3 and D2 receptors, specifically showing the extracellular binding pocket's presence in the D3 receptor, potentially allowing for the development of drugs with decreased side effects for conditions like Parkinson's disease (Arroyo-Urea et al., 2024).



Project Timeline

The Project timeline is as follows:

Makerspace Orientation (including in-person training) Team members: E.W., H.R, T.B., H.D.	08-27-2025
Makerspace 3-D Printer Training (including in-person training) Team members: E.W., H.R, T.B., H.D.	09-05-2025
Project Description and in-class prototype Team members: E.W., H.R, T.B., H.D. Bring ideas to class - E.W., H.R, T.B., H.D. Participate in the in-class prototype - E.W., H.R, T.B., H.D. Project description purpose - E. W. Project description - E. W, T. B, H. D, H.R. Project timeline - T. B. References - E. W, T. B, H.D, H.R.	09-12-2025
Design Brief Team members: E.W., H.R, T.B., H.D. Introduction and Background - E.W., T.B. Project description purpose - E.W. Project description - E.W, T. B, H. D, H.R. Project timeline - T.B. Model description - H.R. Target audience - H.D. Model criteria - E.W., T.B. Design-related research: Existing Resources & Limitations - E.W. Prototype Progress - H.R. References - E.W., H.R., T.B., H.D.	10-03-2025
Draft Design Sketches Team members: E.W., H.D.	10-10-2025

Mid-term Model Design Team members: H.R., T.B., H.D. 3D Printing: E.W.	10-27-2025
Poster Layout/Storyboard Team members: T.B., H.D.	10-31-2025
Poster - First Draft, Revised Mid-term Model Design Team members: E.W., H.R, T.B., H.D. Prototype Pick-Up: H.D. Poster Content: T.B., H.D.	11-07-2025
Poster - Second Draft, Model/Prototype Parts in Class Team members: E.W., H.R, T.B., H.D. Prototype Pick-Up: E.W. Dossier Project Description: H.R. Poster Design & Content: T.B., H.D.	11-21-2025
Makerfest Hosted by UNC BeAM Makerspace Prototype Pick-Up: H.D. Poster Design & Content: H.D., H.R. Team members: H.R, T.B., H.D.	12-03-2025
Final Project (Design Dossier, 3-D Model (photos & videos) and files, Posters, Peer Review Forms) Team members: E.W., H.R, T.B., H.D. Design Dossier: E.W., H.R, T.B., H.D. 3-D Model (photos & videos) and files: E.W., H.R, T.B., H.D. Poster Design & Content: T.B., H.D., H.R.	12-04-2025

The need-date for project completion is 12-04-2025.

Model Description

The model depicts the D3 receptor as a seven-helix membrane bundle with clearly marked extracellular and intracellular faces. On the extracellular side, it shows two binding sockets: a deep orthosteric pocket and a shallower allosteric pocket located around TM1-TM2 and the extracellular loops. A simplified FOB02-04A is represented as

three pieces: the orthosteric head (PP), a linker, and the allosteric head (SP) to demonstrate how each portion fits into its corresponding pocket. The PP derives from a known D3 orthosteric agonist scaffold (PF-592,379 series), so PP-like cores could act as agonists independently. However, in FOB02-04A, the PP is covalently linked to an SP via a chiral cyclopropyl linker; the combined geometry is what yields the observed potency, efficacy, and D3-over-D2 selectivity. On the intracellular face, the model indicates the conformational change that allows for Gi/o binding and then signaling. The earlier deliverable is a static, virtual scene; later iterations add a movable TM6 to physically visualize the inactive-to-active transition once both sites are engaged.

Target Audience

The primary stakeholders for this model are medicinal chemists and molecular pharmacologists, especially those interested in drug discovery. This 3D printed model will provide a tangible, and physical visualization of how a bitopic ligand, in this case FOB02-04A, interacts and engages with the orthosteric and allosteric binding sites of the dopamine D3 receptor. The model will help provide a better understanding of how FOB02-04A binds and its unique mechanism of action for both the orthosteric and allosteric sites (Arroyo-Urea et al., 2024).

Clinicians studying possible treatments for Parkinson's disease may also find value in this model, as other D3 agonists have been shown to potentially alleviate certain symptoms associated with the disease (Carnicella et al., 2014). The model will provide a visual of how the agonist FOB02-04A's binding stabilizes the D3 receptor's active site and will also highlight why bitopic ligands may potentially offer therapeutic benefits with fewer off-target side effects in comparison to traditional drugs used for treatment (Arroyo-Urea et al., 2024).

Both students and educators will also benefit from this 3D model. The complex dual-binding site and conformational changes occurring during the binding of FOB02-04A to the dopamine D3 receptor will be better understood through a 3D model. These concepts may be difficult to understand from 2D diagrams, so this model will offer a more engaging way to demonstrate and study these mechanisms, making the study of the drug and receptor more accessible.

Model Criteria

FUNCTION

The criteria set forth below describe how the proposed model will do what it is intended to do, including move, and should be met to achieve successful completion of the project:

1. The proposed model will be able to move to change between “active” and “inactive” configurations.
2. The proposed model will have two separate “binding sites” to which the orthosteric and allosteric regions of the bitopic ligand FOB02-04A will be able to fit into.
3. Ligand binding to the orthosteric site will cause the model to stay in its active conformation, while ligand binding to the allosteric site will make the orthosteric site more accessible.
4. The model will have associated Gi α , β , and γ proteins. These will respond to conformational change in such a way that the downstream effects of different conformational changes are clear.
5. The model will be similar in shape and structure to the D3 receptor and associated Gi proteins, with binding sites in representative places.

FORM

The criteria set forth below describe how the proposed model will be designed to match its intended function, and should be met to achieve successful completion of the project:

1. The proposed model will be handheld and table-top sized for demos and presentations within class.
2. The ligand orthosteric and allosteric site should consist of detachable components that can be combined by the users to form the full bitopic ligand depending on the conformation shown.
3. The individual ligand pieces will be smaller scaled to ensure that the surfaces and binding pocket will be large enough. The scale will allow ligand insertion or removal to be visible to the readers and viewer.
4. The D3 receptor will be presented as a seven-transmembrane helix bundle with extracellular and intracellular loops. The extracellular face will have an orthosteric and allosteric pocket. The intracellular face will include a platform for Gi α , β , and γ attachment.

5. When using 3D printing, PLA or PETG filaments will be used for the receptor body and base in the model.
6. The model will use consistent color coding to distinguish the orthosteric and allosteric pocket, ligand molecules, $G_i \alpha, \beta$, and γ attachments. Labeling will be engraved or printed on the surfaces.

Model Constraints

The constraints listed below describe design limitations that we must stay within when designing the model to achieve successful completion of the project:

1. Material availability since the model should be fabricated using materials from the UNC BeAM Makerspace (PLA and PETG filaments, printer, slicers, and hand tools). There are no specialized resins or external materials requiring purchase outside of what the instructor suggested and approved.
2. Size and printing volume as the model must fit within the standard print bed dimensions that are available in the UNC BeAM Makerspace. The components can be printed in multiple parts that can be assembled post-printing.
3. Durability as the model should withstand repeated handling without breaking. The model components should minimize safety risks.

Design-Related Research

Existing resources that model our chosen drug and receptor include:

- The UniProt database has a physical structure of the D3 receptor at accession number P35462 (UniProt Consortium, 2025). This database has structures from a variety of techniques, including X-ray crystallography and cryo-EM, as well as computer predicted structure from AI protein structure predictive models such as AlphaFold.
- Primary structural references include
 - Inactive D3R (antagonist-bound): X-ray structure, PDB 3PBL. Used to define the inactive baseline for helix positions and to place the orthosteric pocket in the model.
 - Active D3R (bitopic agonist-bound): Cryo-EM complex, PDB 9F33 (D3R:Go with FOB02-04A). Used to locate the allosteric selectivity site, confirm

dual-site engagement (PP in orthosteric, SP in allosteric), and set the TM6-out displacement used for our hinge/stop.

- Active D3R with orthosteric agonists: Cryo-EM structures with pramipexole/PD128907. Used to communicate that overall active geometry is similar across agonists while FOB02-04A rationalizes D3 over D2 selectivity via the allosteric pocket.
- Previous papers by the Newman research group focus on other potential ligands but still study the binding of D3 receptors. One example of this is in their 2014 paper Beyond small molecule SAR - using the dopamine D3 crystal structure to guide drug design, in which they use a space filling model to show the orthosteric and secondary binding pockets of the D3 model (Keck et al, 2014).
- The paper Structures of the Human Dopamine D3 receptor-Gi complexes provides two cryo-EM structures of the D3 receptor bound to pramipexole and PD128907, two D3 agonists that cause slightly different conformational changes in structure (Xu et al, 2021). This paper provides many views of the two structures, giving a wealth of structural information that can be used to design the model. Additionally, the model gives residue-specific data on the interactions between these dopamine agonists and the dopamine receptor.
- This resource provides computer modelling information on the D3 receptors interactions with a variety of ligands, including the bitopic ligand UNC9994. This is extremely helpful, as bitopic ligands have a novel binding method that is not accurately explored when only the orthosteric or allosteric sites are bound (Lane et al, 2013).

Limitations of these resources include:

- While the UniProt database has many static structures, it only defines the structure of the D3 receptor in the absence of a ligand (UniProt Consortium, 2025). Additionally, there are no structures showing movement or any other structures besides the canonical D3 structure. The models are all ribbon models, which makes visualizing small molecule binding difficult.
- Static figures and database viewers communicate poses well but not the motion or mechanism behind conformational changes. They are unable to convey the interdependence of dual-site contact in a tactile way. Many visuals emphasize the ligand but under-illustrate how the adjustments in position of transmembrane

helices creates the Gi/o docking surface, which is essential for explaining mechanism to non-experts.

- This Newman group paper only has one two dimensional image of the D3 receptor, which makes it difficult to understand the full three dimensional complexity of the receptor. Additionally, the top-down angle of the image does not include how the Gi protein complex is affected by ligand binding (Keck et al, 2014).
- This paper was highly detailed and very useful in terms of the D3 structural changes that occur as a result of ligands binding at an orthosteric site, but lacked any mention of the allosteric binding site. This may make the paper less useful, as the combination of allosteric and orthosteric binding may shift the receptor in a unique conformation unachievable by binding at any one site.
- The views of the bitopic ligand binding are only focused on the ligand itself, and not how the protein as a whole changes confirmation. Additionally, the artistic representation of the binding pocket leaves much to be desired and may not be useful in designing a 3D model (Lane et al, 2013).

Prototype Progress:

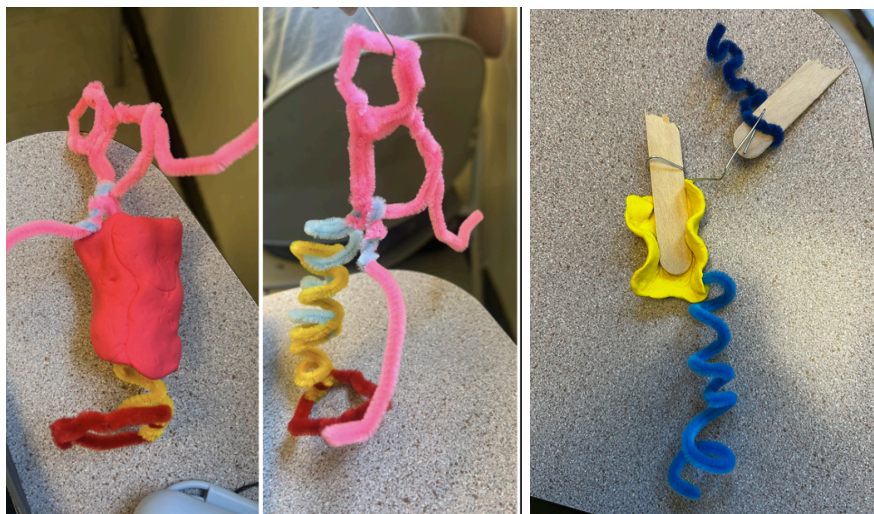


Figure D1. Low-fidelity classroom mock-ups used to communicate the bitopic mechanism. Helices were created out of pipe cleaners to show accuracy to how the D3 receptor might look. The concept for designing the ligand in 3 parts, with two pharmacophores connected by a linker, was developed here.

- Through our initial prototyping, we found that designing the model starting with the helix made structuring the rest of the components confusing in its placement, particularly for demonstrating the target mechanism. We realized our visualization will likely need many moving parts and items to help control the moving parts, since our mechanism involves the ligand engaging two separate binding sites in order for the receptor conformation to move from inactive state to active state. Additionally, we will need to figure out how to make the movements precisely flow into each other.
- Thus instead of starting with building the helices first, which made it harder to communicate the mechanism, we shifted to a site-first approach that defines the orthosteric and allosteric sockets and uses TM6 as the main visual identifier of an inactive vs. active conformation.
- In order to begin 3D modeling on TinkerCAD, we first needed the design of just the dopamine D3 receptor (D3R) in its inactive state. We found a paper that determined the crystal structure of the D3R, purified and stabilized with antagonist eticlopride (Chien et al., 2010). Conducting crystallography with an inverse agonist captures the D3R structure in its “inactive” state (due to being in a conformation that cannot engage a G-protein). The structure of the human dopamine D3 receptor in complex with eticlopride was then taken from the RCSB Protein Data Bank (PDB), and uploaded onto the structure viewer ChimeraX.
- The same was done with the Cryo-EM structure of D3R:Go complex bound to bitopic FOB02-04A - Conformation A.
- In ChimeraX we are able to visualize where the ligands are specifically binding: on both a solid surface, and with the residues and protein structure visible below.

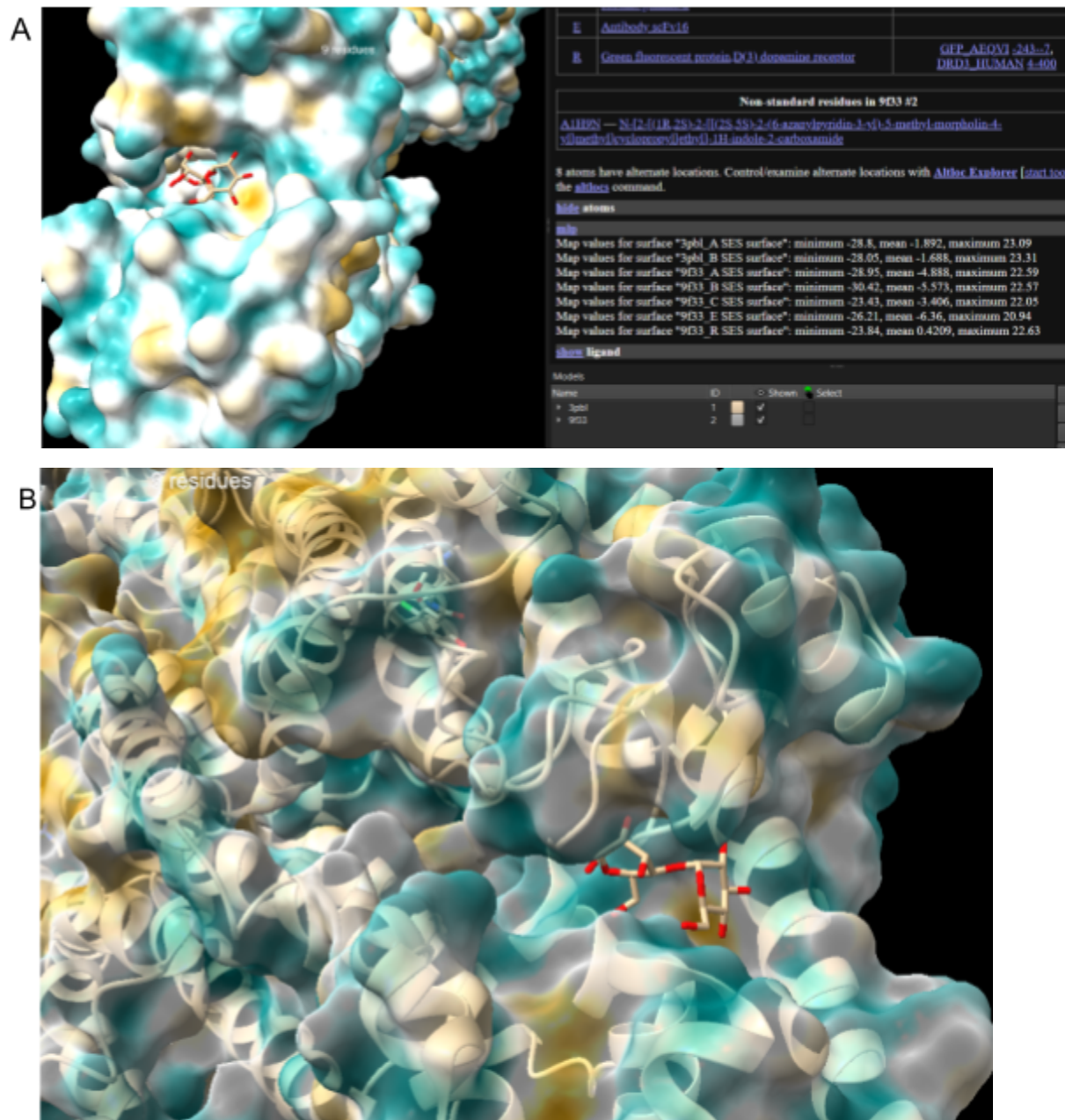


Figure D2. ChimeraX visualizations used to guide pocket placement and verify the bound pose of FOB02-04A in active D3R. (A) Surface rendering of the active D3 receptor from 9F33 (chain R retained), showing the bitopic ligand FOB02-04A (sticks; red/orange) spanning the cavity. The right panel confirms model contents (chain table) and lists the non-standard residue identifier for the ligand. (B) Zoom-in on the binding cavity with the solvent-excluded surface displayed; the primary pharmacophore (PP) sits deep in the orthosteric pocket, while the secondary pharmacophore (SP) extends toward the extracellular vestibule near TM1-TM2/ECL1.

- The next steps are to export surface meshes and the FOB02-04A ligand, carve the two sockets from the active surface in TinkerCAD, and place a three-piece ligand to verify fit.
- Moving forward in designing the model, we must first remove all unnecessary residues, identify which chains to target, isolate those chains, and then align both the inactive and active states to compare the two structures. Once we have verified that the helices (TM1-TM7) overlap reasonably, we should be able to see TM6 visibly tip outward on the internal (cytoplasmic) side.
- We also must demonstrate what exactly our bitopic ligand, FOB02-04A, does to selectively activate the D3 receptor. This can be shown by first constructing the GPCR such that the transmembrane helices TM1-5 and TM7 are kept as a fixed body on the receptor shell, while having a mobile TM6 piece on a pivot in order to act as the hinge that--when opened--allows for the perfect conformation to which Gαoβγ can attach. After which, the downstream effects of D3R activation can be explained.
- The bitopic ligand functionality is best demonstrated in a 3-part system of two pharmacophores joined by a linker. The primary pharmacophore (PP) binds deep in the D3 orthosteric pocket, while the secondary pharmacophore (SP) binds in the allosteric site around TM1-TM2 and the extracellular loops (ECL1-ECL3). It is an extremely minute difference in this site--specifically an extra G94 on the TM2-ECL1-TM1 of D3 and absent in D2--that highlights how the bitopic nature of our ligand allows for subtype selectivity. We must explain that this is necessary because the orthosteric binding site (OBS) on D2 and D3 receptor subtypes are practically identical, and cannot be targeted alone if specificity is desired.
- The biggest anticipated difficulty in design will be getting the linking stem to make both pharmacophore ends bind in the appropriate places at the same time, depicting the bitopic mechanism.

We plan to use the following resources when building our model:

- Structure prep and export in ChimeraX (3PBL chain A; 9F33 chain R).
- CAD in TinkerCAD/Onshape (socket carving with PP/SP bodies; TM6 hinge).
- Printing on UNC BeAM FDM printers (PLA/PETG).



Design Sketches

As a bitopic ligand, our drug of interest has two regions that each bind to a different pocket of the D3 receptor. To model this, we will 3D print two distinct ligand heads, one for the orthosteric (deep) pocket and one for the allosteric (shallower) pocket. The flexible linker that connects the two regions will be modelled by a pipe cleaner or similar flexible piece.

The D3 receptor will be a cylindrical shape, with shallow grooving in the side representing the 7 transmembrane helixes characteristic of GPCRs, with TM6 clearly labeled as the “activation lever.” This will consist of two equal halves, which will be held together by duct tape or something similar on the bottom and rubber bands attached to pins on the sides. This will allow the model to hinge open and expose the allosteric site, and will allow it to close to represent the inactive state. The energy required to open the receptor against the rubber bands represents the energy required to change conformational states. The model will be staged so that dual-site engagement (orthosteric and allosteric) corresponds to the active pose, which is in line with the bitopic mechanism we are illustrating.

The binding pocket is designed such that the agonist can bind on the orthosteric pocket, but can easily be removed, unable to fully stabilize the active state. However, the two heads together will stabilize the active conformation. The positive allosteric modulator when combined with the ligand serves to lock the ligand in place, as it requires the model to open up to release the ligand. This simulates the greater binding affinity of the bitopic ligand as compared to the agonist alone.

One drawback with this model is that the binding pocket is oversized as compared to the entire receptor. This is intentional, as it places a bigger focus on the binding pocket and helps to offset the limitations created by the relatively small size of the receptor. However, care should be taken to ensure that this model is not misrepresented as a scale model of the D3 receptor.

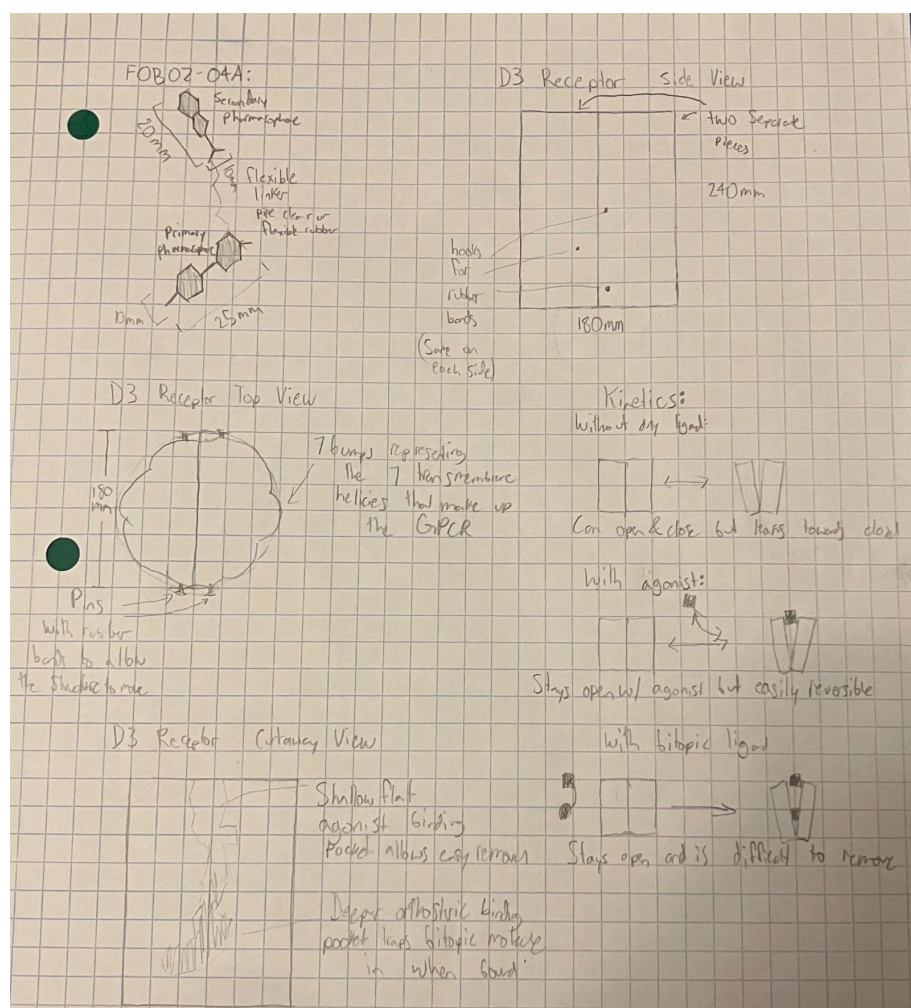


Figure S1. Inactive D3R exterior: helical layout and pocket landmarks. Pencil sketch of the seven-helix bundle with TM1-TM7 indicated as surface grooves; TM6 is labeled for later comparison. Surface landmarks identify the central orthosteric pocket and the shallower, off-center vestibular site near the extracellular rim. Dimensions at model scale and a 20-mm scale bar are annotated.

Revised Design Sketches

The following figures are revised sketches illustrating the same core bitopic mechanism presented initially, but has been made clearer and more organized. Our model will demonstrate a bitopic ligand with two units connected by a flexible linker engaging both the deeper orthosteric site, and the shallow allosteric pocket of the D3 receptor. When bound, the bitopic ligand will stabilize an active conformation of the receptor. While there has been no change to the concept of what we are doing, the updated figures provide improved visuals that are more clear to interpretation.

FOB02-04A

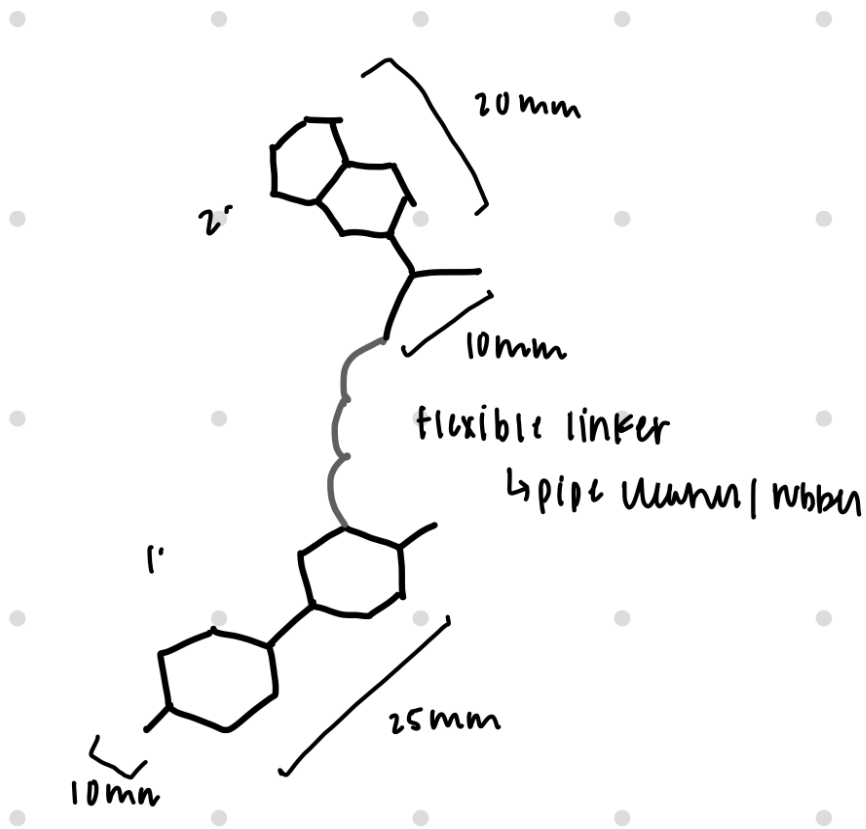


Figure RS1. FOB02-04A bitopic ligand schematic. The figure above shows a ligand consisting of primary and secondary pharmacophores connected by a flexible linker (~10mm). The linker will allow for simultaneous engagement of both the orthosteric and allosteric sites. The primary pharmacophore will anchor deeper into the orthosteric pocket, while the secondary pharmacophore will interact with the superficial allosteric pocket.

D3 Receptor

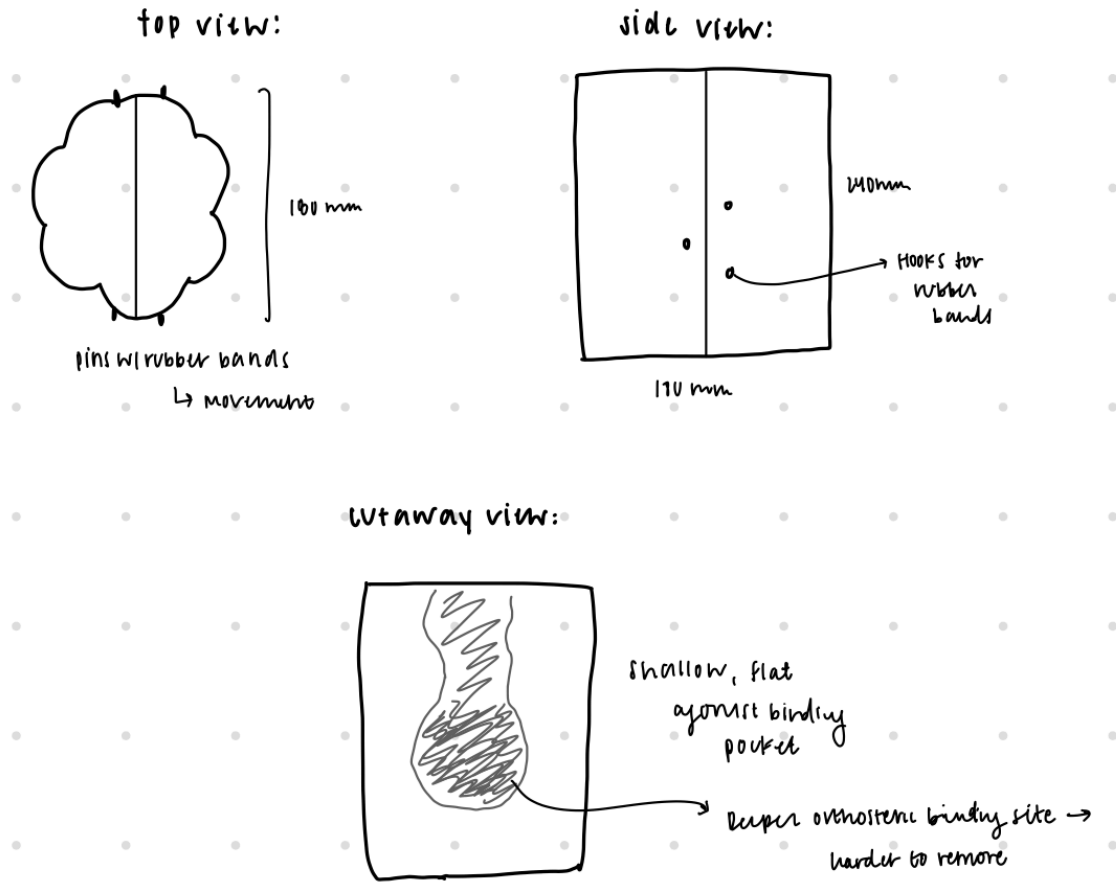
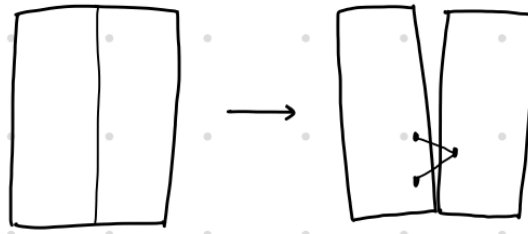


Figure RS2. D3R model geometry and binding pocket layout. The top view shows two connected halves of the cylindrical receptor, showing the closed, inactive state. The side view explains how the model will be able to hinge open to expose the binding sites. The cutaway view highlights the deep orthosteric site, where the primary pharmacore will bind, and the shallower allosteric site, where the secondary pharmacore will bind.

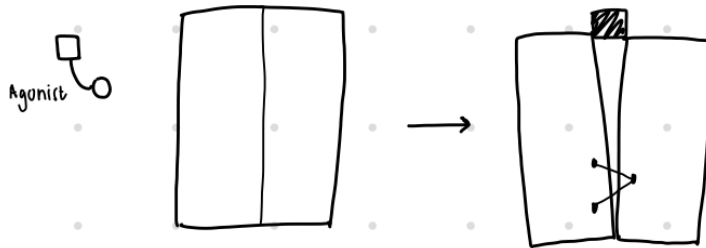
Kinetics

No ligand:



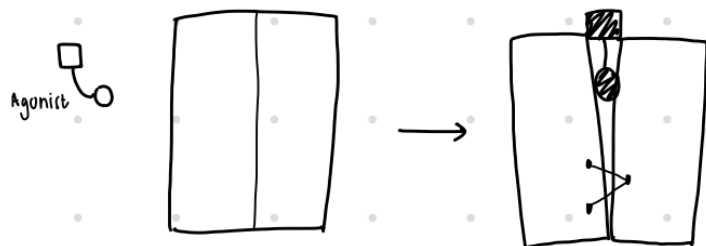
opens + closes; not fully opening

Agonist bound:



very easily reversible
but will stay
open

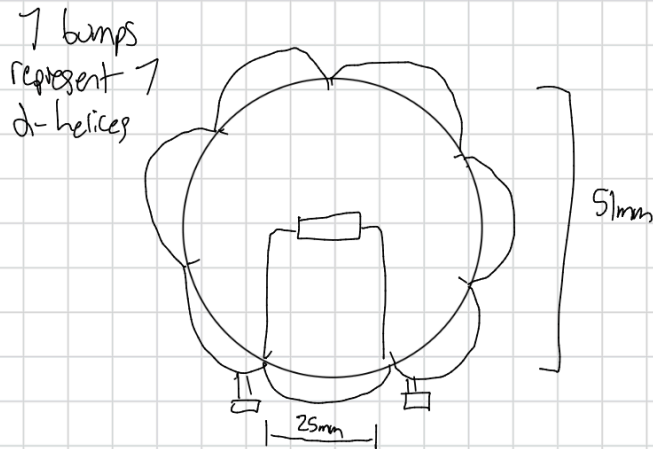
Bitopic ligand bound:



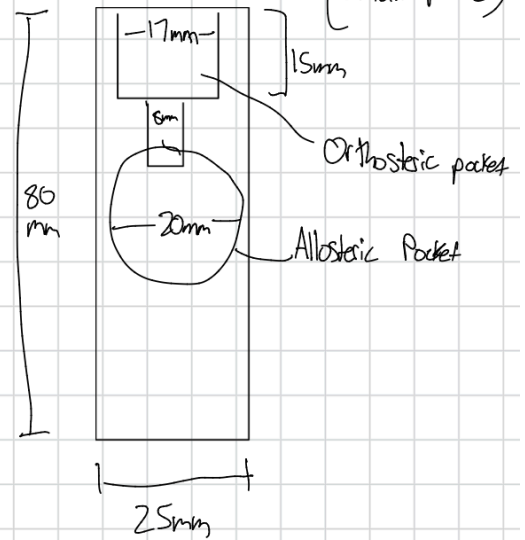
ligand hard to remove +
stays open more

Figure RS3. Model kinetics. In the state where no ligand is present, the model opens and closes freely, but remains mainly in the inactive conformation. When an agonist binds, the receptor opens but remains weakly stabilized. The ligand can, at this point, be removed easily. When a bitopic ligand binds, the primary pharmacore anchors to the orthosteric site, and the secondary pharmacore engages with the allosteric site, thus stabilizing the open conformation. The ligand is now harder to remove and the receptor will stay open.

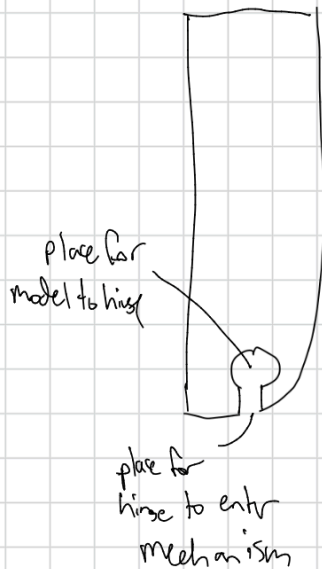
D3 Receptor top View:



D3 Receptor Inside view (Small piece)



D3 Receptor side view (Small piece)



The revised design sketches above are associated with the final, revised 3D model and provide a better depiction of the overall structure of the D3R and the conformational change that occurs when the bitopic ligand binds. The model will be redesigned so that only the α -helix, TM6, moves, while the remaining α -helices remain fixed. The small piece serves as TM6. The earlier sketches showed an oversimplified mechanism that did not show this conformational change as well.

Prototyping

10/26 Prototype

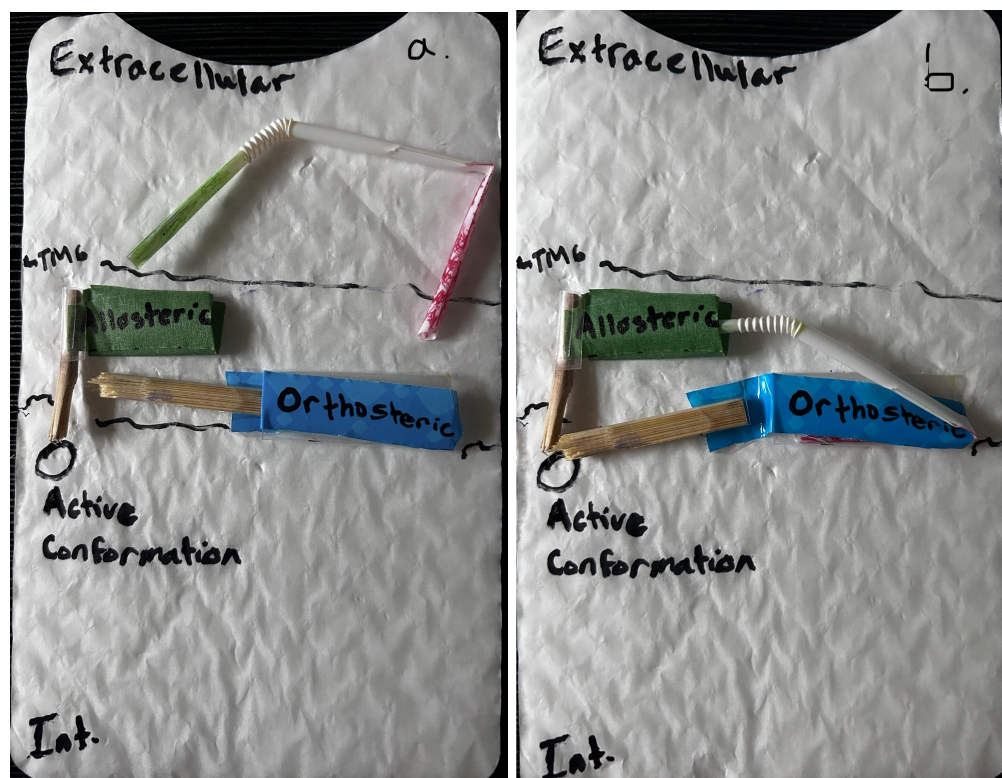


Figure P1. Mockup Model of Pocket Placement. Straw represents the bitopic ligand of interest. Red side of straw represents the primary pharmacophore (PP) that will bind at the orthosteric site. Green side of straw represents the secondary pharmacophore (SP) that will bind at the allosteric site. Both sites need to be engaged to activate the D3 receptor, visually represented with a styrofoam board background. (a) Inactive conformation of D3 receptor. (b) Active conformation of D3 receptor.

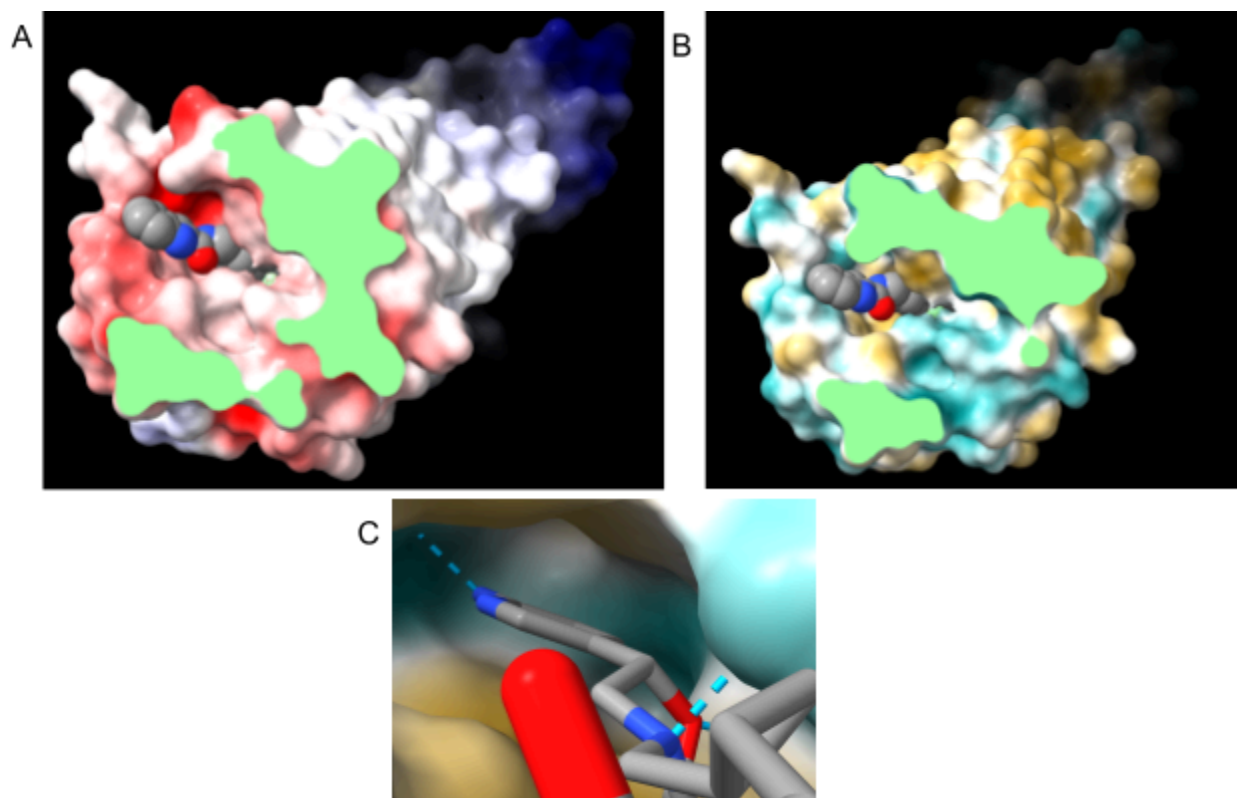


Figure P2. 3D model showing ligand interactions on an accurate D3R depiction. (A) Extracellular view of active D3R with surface electrostatic coloring (red: areas with high electron density/negative potential, attracting positive ions; blue: areas with low electron density/positive potential, attracting negative ions) and the bitopic ligand in atomic sphere form. (B) Extracellular view of active D3R with surface hydrophilicity coloring (blue: hydrophilic; orange: hydrophobic) and the bitopic ligand in atomic sphere form. (C) Closeup of hydrogen bonds between the bitopic ligand in stick representation and its binding sites. This surface representation of the D3 receptor was further edited in Blender to make it suitable for 3D printing, but final printing iterations on this more structurally accurate model ended up failing and returning printing errors. Thus it now serves as a 3D virtual model for giving an accurate visualization of how the receptor and ligand actually look and interact for those audiences who want to see more beyond the simpler breakdown of the mechanism from our 3D printed model.

11/7 Prototype:

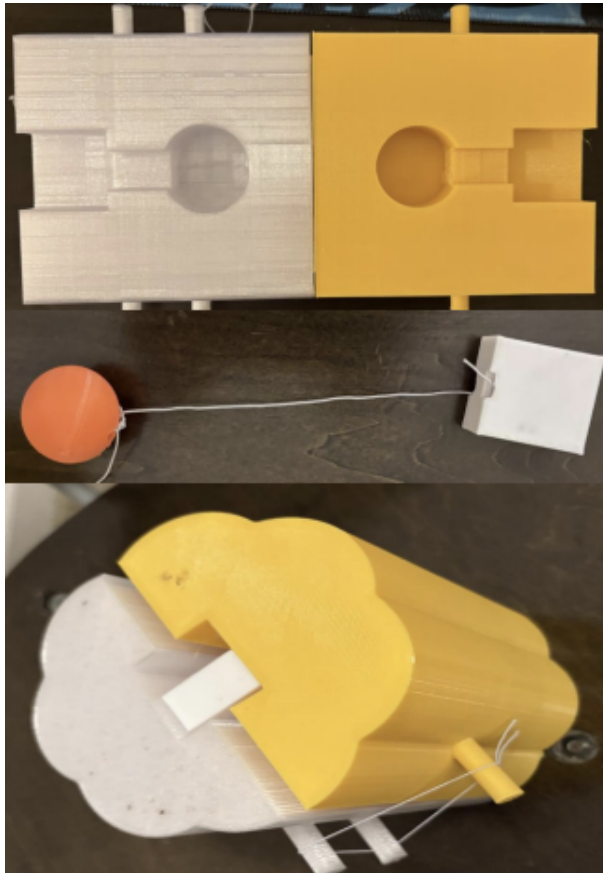
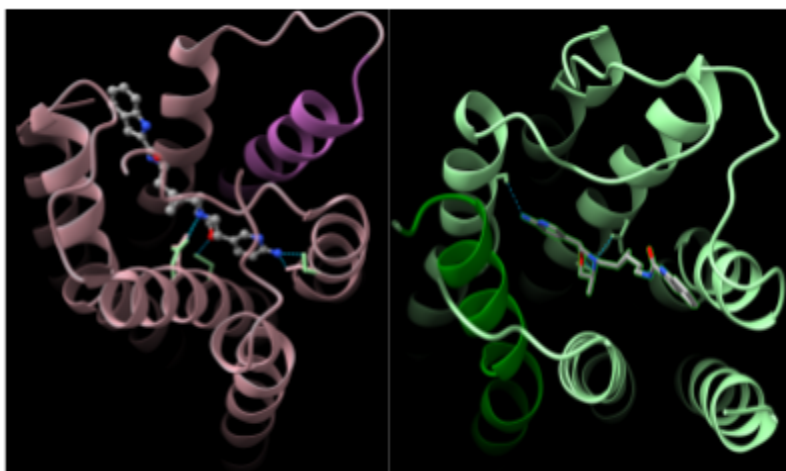


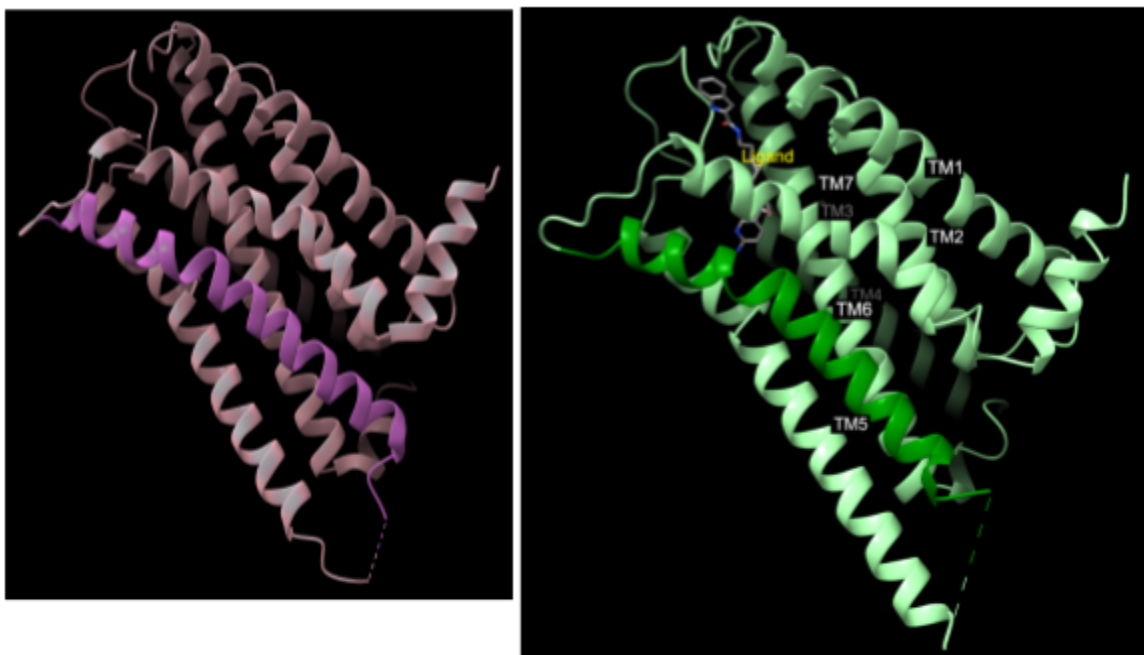
Figure P3. Initial 3D Printed Model. The initial model consisted of a two piece receptor that was held together with rubber bands and tape to allow the model to open and close. The model is symmetrical and opens directly down the middle. The model provides an overly simplified activation mechanism where when the bitopic ligand binds to the orthosteric and allosteric sites, the model “locks” better than if a normal ligand were to bind. The model however, did not stabilize as well as needed. The two parts of the ligand are bound together with a rubber band.

11/20 Prototype:

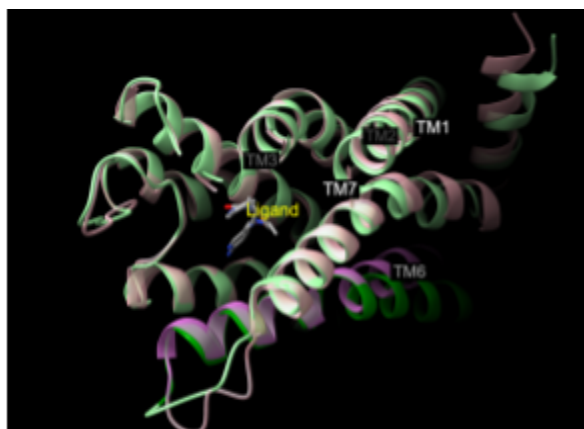
A



B



C



D

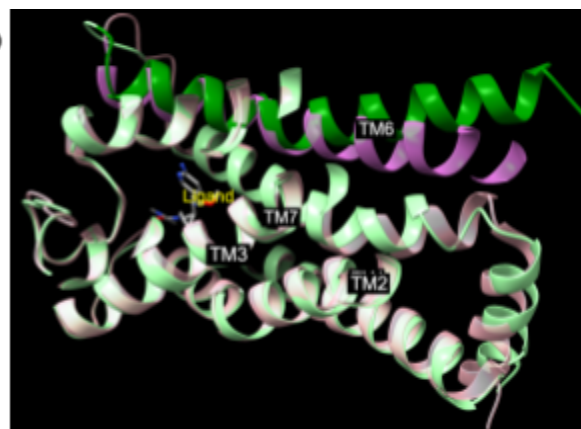


Figure P4. Cartoon representations highlighting accurate D3R conformational changes.

(A) Extracellular perspective of the D3R inactive (pink) and active (green) conformations in ribbon form and bitopic ligand H-bonding to visible residues of what would be the active conformation. (B) Left: Side view of the inactive D3 receptor, with TM6 visible in the front. Right: Side view of the bitopic ligand and the active D3 receptor, with TM6 visible in the front. (C) Extracellular view of the bitopic ligand alongside the D3 receptor inactive and active conformations overlaid on each other. (D) Side view of the bitopic ligand alongside the D3 inactive (pink) and active (green) receptors, better demonstrating the conformational changes upon FOB02-04A binding. TM6 is deeper colored than the other transmembrane helices to show that its displacement is one of the biggest indicators that the receptor went from inactive to active, as the active conformation allows for the intracellular G-proteins to attach and begin the signaling cascade.

11/21 Prototype:

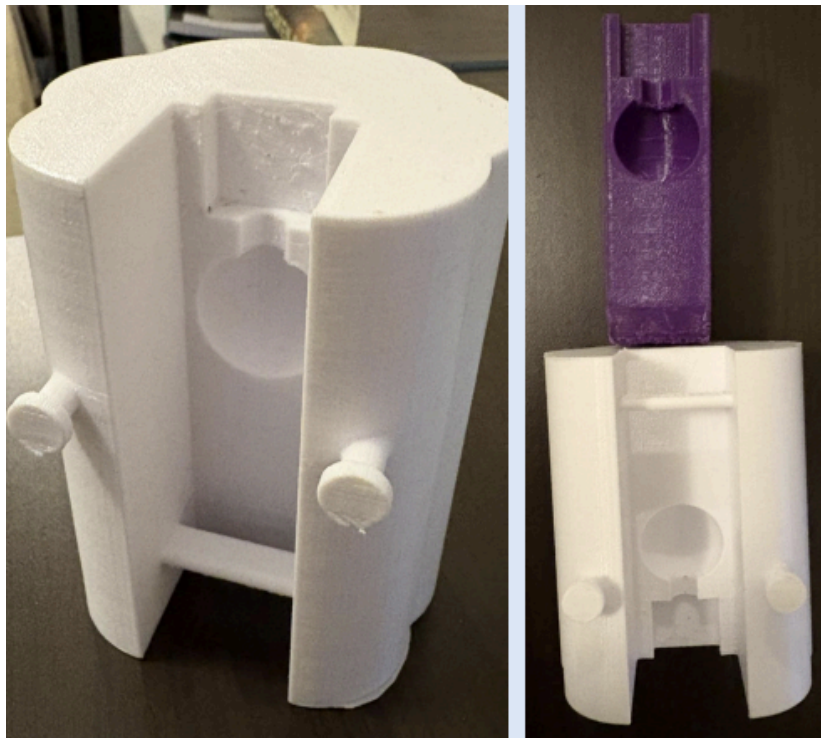


Figure P5. Revised 3D Model. The revised model above consists of a two piece receptor that is no longer symmetrical. The model provides a better depiction of the conformational change occurring at TM6 when the D3R is activated by the bitopic ligand. The model has (a) one alpha-helix, TM6, that opens and closes, while the (b) remaining helices stay fixed. The model utilizes a hinge mechanism, held together by rubber bands in

the inactive state. When the bitopic ligand is bound, the receptor should show the conformational change occurring as the receptor goes into the active state.

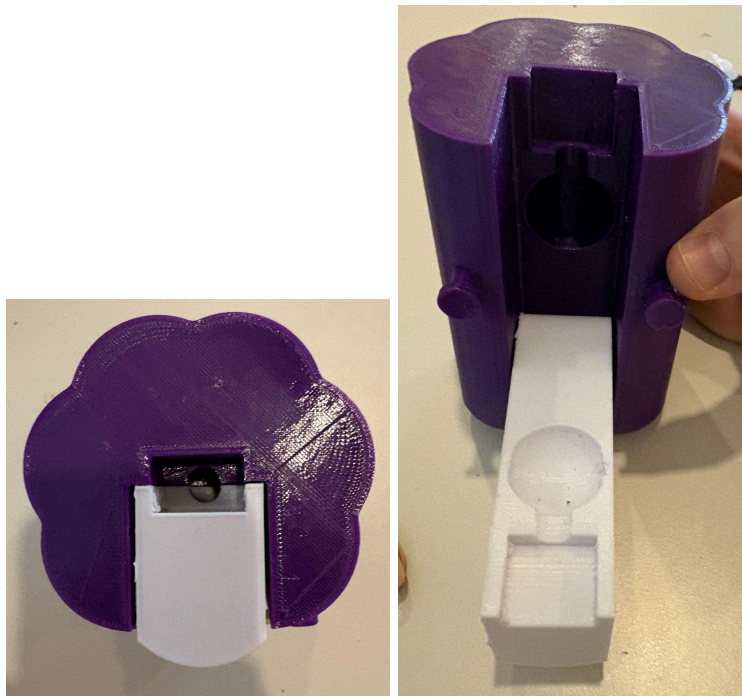


Figure P6. Revised 3D Model. This model is almost exactly the same as Figure P3, with the only change being the dimensions of the smaller component. Figure P3 printed the larger and smaller component with 0 offset, which resulted in the two components not fitting together. Figure P4 made the smaller component slightly narrower, allowing for the hinge mechanism to function properly.



Design Changes and Improvements

As of 10-26-2025, the following revisions have been made to the design submitted on 10-27-2025:

- More simplified mechanism representation as the bitopic ligand mechanism was simplified to its essential components for maximal clarity. The model focuses on a basic interaction between the orthosteric and allosteric sites without secondary structural complexities. This change would ensure the model is accessible and easy to understand while conveying the core concept of bitopic binding and receptor activation.

- Planned incremental detailing as future interaction of the model will build upon this simplified version by adding accuracy, additional labeling, and potentially elements to show conformational transitions more precisely. This stepwise approach will allow for continuous testing of clarity and functionality.

As of 10-29-2025, the following revisions have been made to the design submitted on 10-27-2025:

3D Model with Ligand Interactions on an Accurate D3R Visualization

- Generated a higher-fidelity virtual model of active D3R with the bound bitopic ligand (FOB02-04A) to verify pocket geometry and the dual-site pose before printing.
- Electrostatic surface confirms deep orthosteric seating and surface exposure toward the extracellular vestibule. Hydrophilicity map (blue = hydrophilic; orange = hydrophobic) supports a shallow vestibular contour adjacent to the ligand's SP.
- Earlier mesh-only prints lacked clear guidance for socket depths and clearances. Close-up of H-bond contacts in stick form validates the PP and SP anchoring points used for socket placement. Virtual validation provided precise depth/width targets and showed where clashes could occur.
- Attempted to export this high-detail surface for printing; printer returned non-manifold errors. We therefore designated this asset as a virtual teaching model (for posters/screens) and used simplified, watertight shells for physical sockets.

As of 11-7-2025, the following revisions have been made to the design submitted on 10-27-2025:

Initial 3D Printed Model Design

- The initial model was designed in OnShape based on previously drawn sketches. It was printed out and worked as designed
- A lack of 3 dimensional sketches for the ligand led to a mis-sized orthosteric ligand that did not function to open the receptor as intended.
- The halves of the model opening up were not representative of the actual conformational changes occurring in the D3 receptor in response to FOB02-04A ligand binding.

As of 11-20-2025, the following revisions have been made to the design submitted on 12-05-2025:

Cartoon Representations highlighting D3R Conformational Changes

- Replaced cluttered surface renders with clean ribbon cartoons of inactive (pink) and active (green) D3R to communicate TM6 displacement and ligand seating more clearly. Stakeholder feedback: surfaces were visually heavy; helices and the ligand path were hard to follow. Cartoon views improved legibility for live explanation.
- Inactive/active overlays show the TM6-out hallmark and how it exposes the intracellular coupling face.
- Prepared final labels (TM numbers, Extracellular/Intracellular, Orthosteric/Allosteric) to engrave/paint on prints.

As of 11-21-2025, the following revisions have been made to the design submitted on 10-27-2025:

Revised 3D Modeling Design

- Integrating feedback from team members and peer review, another 3D model was designed from the ground up.
- This model kept the simple geometry of the previous model while changing the kinetic element to more accurately reflect the motion of the D3 receptor conformational changes.
- In this model, only one alpha helix of the D3 receptor moves in response to ligand binding. This is more realistic, as the majority of conformational change in response to F0B02-04A binding occurs at the sixth alpha helix.
- A hinge mechanism was added so tape was no longer necessary.
- The smaller piece of the model printed with no offset, resulting in the two parts of the receptor not fitting correctly, making the hinge mechanism not work

As of 11-22-2025, the following revisions have been made to the design submitted on 12-05-2025:

Final, Revised 3D Model Design

- The smaller piece of the model was reprinted to fix the last error with the offset, allowing the two components of the receptor to fit together, and act as a hinge.
- All other components of the model were kept the same.



Materials, Tools, and Equipment

We will use the following materials, tools, and equipment to construct our model (as of 10/27):

MATERIALS

1. Tape
2. Colored paper
3. Styrofoam
4. Plastic straw
5. Chopsticks
6. PLA or PETG filament (receptor body, TM6 arm, ligand)
7. Elastic bands
8. Printed pegs (anchors/axle)
9. Contact adhesive (post-print assembly)
10. Paint and Pens

TOOLS

1. Scissors
2. Pliers
3. Sharpie
4. Colored markers

EQUIPMENT

1. ChimeraX
2. Blender
3. TinkerCAD/Onshape
4. Slicer
5. UNC BeAM FDM printers

For 3D printing, the Ultimaker 3 will be the primary machine used to print PETG plastic parts. These 3D printed PETG plastic parts will make up the bulk of the receptor model and the two pharmacophores of the ligand model.

In the future, rubber bands will be used to provide the flexible linker between the two pharmacophores of the ligand and to bring the two halves of the receptor model together. On the underneath of the receptor model, tape will be used to permanently attach the two halves and act as a hinge.

The initial and revised design sketches were made in Goodnotes. The 3D designs were made in Onshape, and exported to 3dPrinterOS for support generation and printing.

Final Model

Our final model uses the same general concept as our initial 3D printed model but with refinements to better represent the conformational changes the D3 receptor undergoes as a result of its binding with the FOB02-04A receptor.

Final 3D Printed Model:

Onshape Document (Online 3D Modeling Software):

<https://cad.onshape.com/documents/faedf1b3e9cdaac4dac6b15e/w/441c22595c05f2daa720b04c/e/950ec2b1f99f807973e8783d?renderMode=0&uiState=69324a9ea308cf374b32576d>

Video of the model in its different conformations:

https://drive.google.com/file/d/1yZ_c30h6e0LN8_3JjLUjyJ_34UDwPcA9/view?usp=sharing

Video of the model interacting with the orthosteric and bitopic ligands:

https://drive.google.com/file/d/1C_DjaqikZIGduULoQ2THBfllO36VggCg/view?usp=sharing

Images:

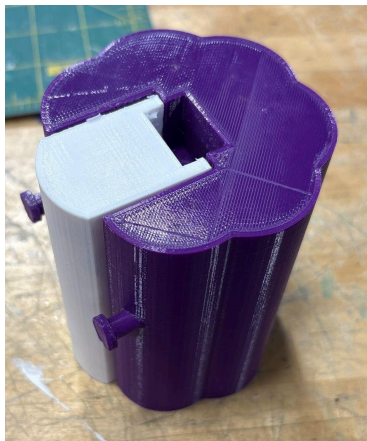


Image 1. Three-quart view of the finished D3 receptor model

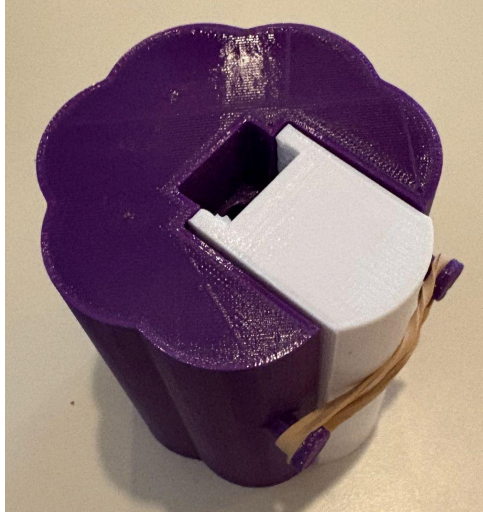


Image 2. Another three-quarters view of the finished D3 receptor model



Image 3. Side view of the D3 receptor



Image 4. Top view of the D3 receptor. The space in the middle show above is the orthosteric binding site.



Image 5. An inside look at the D3 receptor model. The round pocket is the allosteric binding site, while the rectangular pocket is the orthosteric binding site.



Image 6. A cardboard model of the orthosteric ligand, representing a traditional D3 receptor agonist such as dopamine



Image 7. A cardboard model of the bitopic ligand FOB02-04A, which has both allosteric and orthosteric pharmacophores. The allosteric pharmacophore is shown made of cardboard on the left side, while the cardboard orthosteric pharmacophore is on the right. The linker is made of pipe cleaners.



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Contributions

T.B. [Project timeline, Project description, Model Criteria, Materials/Tools/Equipment, Model Constraints, References, Design Changes, Poster Design, Poster Content, Poster Editing, Design Changes and Improvements, Design Dossier Editing].

E.W. [Project description purpose, Project description, Model Description, Model Criteria, Design Related Research, References, Design Sketches, Materials/Tools/Equipment, Poster Information, 3D Prototyping, 3D Model Design, 3D Printing].

H.D. [Project description, Target Audience, Materials/Tools/Equipment, References, Revised Model Sketches, Poster Content + Design, Prototyping (Dossier), Design Changes and Improvements].

H.R. [Project description, Project description purpose, Model Description, Design-Related Research, Materials/Tools/Equipment, References, 3D Prototyping, 3D Model Design, Video Visualizations, Design Changes and Improvements, Poster Content, Poster Editing, Design Dossier Editing].